Customer Shopping Behavior & Analytics

Executive Performance Report & Data Pipeline Architecture

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INSTITUTION
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PIPELINE
(Python+SQL + Power BI)

1. EXECUTIVE SUMMARY

This project delivers an end-to-end data analytics and business intelligence solution analyzing consumer purchasing trends, customer segmentation, and product performance. By integrating Python-driven preprocessing, MS SQL Server analytical views, and an interactive Power BI dashboard, leadership gains direct visibility into high-value revenue drivers and retention levers.

2. KEY PERFORMANCE INDICATORS

TOTAL CUSTOMERS	AVG SPEND / ORDER	AVG SATISFACTION	SUBSCRIBED USERS
3,900	$59.76	3.75 / 5	27%
	Optimal price point	Consistent quality	Key retention vector

3. TECHNICAL ARCHITECTURE OVERVIEW

The analytics pipeline transforms raw consumer records into actionable executive insights through a three-stage modern data architecture:

Pipeline Workflow:
1. **Python / Pandas:** Missing data handling (category-median strategy), feature engineering (age binning, purchase frequency calculations), and database delivery.
2. **MS SQL Server:** 11 pre-computed analytical views enforcing consistency and optimizing query speeds for BI consumption.
3. **Power BI & DAX:** Direct relational connection powering interactive filtering, demographic slices, and key performance metrics.

4. MS SQL SERVER ANALYTICAL VIEWS

To maintain high performance and prevent heavy client-side computation in Power BI, the project establishes a modular relational view layer in MS SQL Server (`Quries/Customer_DB_Querirs.sql`).

SQL VIEW NAME	ANALYTICAL PURPOSE & STRATEGIC FOCUS
`vw_customer_behavior_base`	Master sanitized view containing core customer traits and transaction values.
`vw_gender_revenue`	Aggregates gross revenue and total order volumes by gender classification.
`vw_subscription_summary`	Compares spending behavior between subscribed vs non-subscribed customers.
`vw_customer_segment_count`	Segments user cohorts into *New*, *Returning*, and *Loyal* tiers based on order counts.
`vw_top3_items_per_category`	Uses window functions (`ROW_NUMBER`) to isolate top product revenue drivers.
`vw_discount_above_avg`	Monitors high-value transactions that utilized above-average promo discounts.
`vw_shipping_avg_purchase`	Evaluates customer basket size variations across different shipping speeds.
`vw_age_group_revenue`	Calculates gross lifetime value and spend density across age brackets.

5. DATA PREPROCESSING & PIPELINE INTEGRITY

Data quality was prioritized prior to SQL ingestion. Using Python (`Python_Script/`), missing values were resolved using robust category-median imputation rather than simple mean substitution to prevent outlier skewing. Feature engineering introduced two key variables:

- **Age Grouping:** Binned raw ages into targeted demographic cohorts (Youth, Young Adult, Adult, Senior) for localized marketing analysis.
- **Purchase Frequency Index:** Standardized transaction intervals to identify customer repurchasing cycles.
- **Redundancy Reduction:** Removed correlated or static feature flags to maintain clean database schema standards.

6. BEHAVIORAL FINDINGS & MARKET INSIGHTS

27%	$59.76	3.75 ⭐
SUBSCRIPTION ADOPTION	MEAN ORDER VALUE	REVIEW BENCHMARK

Comprehensive analysis across demographic segments and purchasing histories revealed critical behavioral patterns regarding customer spend, promotional sensitivity, and brand loyalty:

- **Subscription Premium:** Subscribed members demonstrate a 15% higher purchase frequency than non-subscribers, representing the primary driver for predictable recurring revenue.
- **Category Dominance:** Clothing and Accessories represent the largest revenue contribution across all age cohorts, with Young Adults exhibiting the highest average order value (AOV).
- **Shipping Mode Sensitivity:** Customers choosing Express or Next-Day shipping show a higher average purchase value ($62+) compared to standard shipping choices.
- **Discount Usage Pattern:** High-value transactions show moderate sensitivity to discounts, indicating that promotions effectively increase basket size without degrading brand perception.

7. STRATEGIC BUSINESS RECOMMENDATIONS

- **Accelerate Subscription Incentives:** Introduce targeted post-purchase discounts to transition one-time buyers into the 27% subscriber tier.
- **Optimize Express Shipping Perks:** Offer complimentary express shipping thresholds for orders exceeding $75 to capture higher margin carts.
- **Targeted Demographic Campaigning:** Align marketing spend toward the high-performing Young Adult demographic in core categories like Apparel and Footwear.